Mardente, John <FRA>

From: Cothen, Grady <FRA>
Sent: Tuesday, May 29, 2007 9:32 AM
To: Reynolds, John <FRA>; Myers, David <FRA>
Cc: Mardente, John <FRA>
Subject: FW: John Reynold's Recommendation at Farnsworth ave

Please advise.

Grady C. Cothen, Jr.
Deputy Associate Administrator
for Safety Standards and Program Development
Federal Railroad Administration
202-493-6302
240-687-4355 (cell)

From: JNorth@njtransit.com [mailto:JNorth@njtransit.com]
Sent: Friday, May 25, 2007 2:46 PM
To: Cothen, Grady <FRA>
Subject: RE: John Reynold's Recommendation at Farnsworth ave

Grady,

This morning I rode a test train with Bombardier management. This train was operated in non-revenue service. We left Bordentown Station going northbound. The LRV Operator was instructed to proceed slowly and stop the train when he had a full unobstructed view of the Farnsworth Avenue crossing. Bombardier management employees then measured the distance from the LRV to the Farnsworth Avenue crossing. The distance was 250 feet.

I compared that site distance to the Emergency Brake Rate Stopping Distance chart on page 9 of our May 12, 2005 waiver request. The stopping distances indicated in this chart are from Bombardier Test Report PS31-1346-39, Appendix 2 - Emergency Brake Care @3502; and Bombardier Test Report - PS31-1346-39. These test reports indicate that at 20mph the River LINE light rail vehicle stops in 67 feet using emergency braking and at 30mph full stop is achieved in 132 feet of emergency braking. Extrapolating the results for 25mph produces a full stop braking distance of approximately 100 feet.

25mph is currently the maximum authorized speed for trains moving over the Farnsworth Crossing. NJT and Bombardier have agreed that maintaining 25mph as the maximum authorized speed provides adequate stopping distance in the event the crossing is blocked.

Additionally; we will do tree trimming on the curve this weekend to improve the site lines beyond the existing 250 feet.

Maintaining 25mph eliminates the complexity you reference below. Do you find this acceptable?

Regards,

Joe North

From: grady.cothen@dot.gov [mailto:grady.cothen@dot.gov]
Sent: Thursday, May 24, 2007 1:00 PM

To: North, Joe F. (CLROJFN)
Subject: FW: John Reynold's Recommendation at Farnsworth ave

Joe, our Region agrees that the marks were likely from the crane. However, it seems to never stop. See below.

A suggested compromise would be to reduce speed from the current 25 to a speed permitting the train to stop short of the crossing upon establishing sight view, if any vehicle is hung up or stored on the crossing, but not to exceed 20 mph when traversing the crossing in any case. (I can't wait to see the rule book test question on that one.)

That might work out to a straight 20 mph, depending on the sight distance and stopping distance. Don't know. Allow for operator reaction time when calculating.

I won't propose restricted speed, because that is just a liability trap at a crossing, given the propensity for last-second things to happen.

I don't want to propose that to the mayor without your agreement, because it will affect schedule by a few seconds on a line with plenty of stops. Plus the added complexity, which is never good.

What's your pleasure?

Grady C. Cothen, Jr.
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From our Field Investigation:

Farnsworth Ave.

Tuesday May 22, 2007 I (again) interviewed the owner of the sound studio located off Farnsworth Ave, Bordentown, NJ. He stated to me that he noticed "NO" interference from the train sounding the horn in the Northern Direction. He also stated that he did see vehicles stuck on the roadway crown. He noted a 1969 Red Chevy Impala convertible (classic) trailering a boat last year stuck solid and personnel from the yacht club assisted him to clear the crossing. He also said he noticed a Beer / Beverage truck hung up on the crossing last year. And finally a municipal trash hauler was also stuck. The real issue is not the sound studio this time but rather the homes (expensive) along the route where the horn is now sounded.